Dark Web Threat Intelligence: Monitoring Cybercriminal Activities and Emerging Cyber Threats*

Abdullah Abid Ali[1]

Abstract—The dark web has become a key centre of coordination of cybercriminals, illegal trade, and propagation of sophisticated cyber threats. In this paper, the researcher provides an extensive analysis of the threat intelligence of the dark web (DWTI), its infrastructure, the type of cybercrime that is done on the dark web and how to monitor and analyze emerging threats. The most critical types of threats such as ransomware-as-a-service, zero-day marketplaces, and data breach ecosystems are addressed in relation to well-known threat intelligence models. Proactive monitoring defensive tools and strategies are described, and the aim is to provide security practitioners with actionable intelligence in order to defend their organizations.

I. INTRODUCTION

In the way that it is most often used, the internet is merely a segment of the entire network infrastructure that the users can utilize. The surface web is the top layer, followed by the deep web, which is content not indexed by default search engines, and the deep web contains the dark web, a set of overlay networks, only accessible by use of special software like The Onion Router (Tor) or the Invisible Internet Project (I2P). In contrast to the deep web, which also includes harmless resources (university databases and company intranets), the dark web has been linked with more and more illegal and criminal activity (such as cybercrime markets, selling stolen data, and organizing attacks by nation-states).

The role of the dark web in the current cybersecurity cannot be overemphasized. Recent industry reports have indicated that a significant percentage of threat actors utilize dark web sites to purchase, sell and trade tools and stolen credentials. This leaves Security Operations Centers (SOCs) and threat intelligence teams with little choice, but to broaden their monitoring posture beyond the traditional perimeter defenses, into dark web surveillance This paper aims at:

- Elaborating the structure and ecosystem of dark web.
- Determining types of cybercriminal acts in the dark web.
- Explaining possible threat intelligence frameworks that can be applied to DWTI.
- Disclosing the techniques and technologies that are used to monitor dark web threats.

Proposing defensive strategies (proposed on the basis of DWTI).

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II. DARK WEB ECOSYSTEM AND INFRASTRUCTURE

A. Layers of the Internet

There are three layers of internet that are often conceptualized. The surface web is the web that is publicly accessible, indexed and is accessible using traditional browsers and search engines. The deep web contains material not indexed, including password-protected databases, bank portals, and medical records, and is estimated to be many orders of magnitude bigger than the surface web. The dark web is a consciously anonymized part of the deep web, only available through special anonymization networks.

B. Anonymization Technologies

The main technologies which allow access to dark web include:

Tor (The Onion Router): Tor directs user traffic by making a path over a sequence of volunteer-provided relay nodes, encrypting each communication layer at every hop. The high level of anonymity is achieved in this multi-layered encryption because a relay does not know the source or the destination of traffic. Tor also contains onion sites, or hidden services, which entirely use the Tor network.

I2P (Invisible Internet Project): I2P implements a garlic routing protocol which encrypts several messages and groups them together. In the dark web forums and file sharing, unlike Tor, which is optimized to work with external sites anonymously, I2P is optimized to work within a network, so it is made to be popular.

Freenet: A peer-to-peer network dedicated to censorshipresistant storage and communication of data. Freenet archives information in a decentralized manner and therefore retrieving and attributing the information is incredibly hard by law enforcement agencies.

C. Dark Web Marketplaces and Forums

Dark web markets operate in a similar manner as online shopping websites, which allow the exchange of illegal goods and services. In Table I, we identify the prominent dark web markets and their features.

[1] Abdullah Abid Aliis a student of Punjab University College of Information Technology, abdullahmian714@gmail.com at papercept.net

Out of marketplaces, there are cybercriminal forums, including RaidForums, XSS.is and Exploit.in, which act as both knowledge sharing and recruiting platforms.

TABLE I

NOTABLE DARK WEB MARKETPLACES

Marketplace	Active Period	Primary Activities
Silk Road	2011–2013	Drug trade, counterfeit documents
AlphaBay	2014–2017	Drugs, malware, stolen data
Hansa Market	2015–2017	Drugs, cybercrime tools
Dream Market	2013–2019	Drugs, carding, exploit kits
Hydra	2015–2022	Drugs, money laundering
BreachForums	2022–2023	Stolen databases, credentials

III. CYBERCRIMINAL ACTIVITIES ON THE DARK WEB

A. Historical Perspective

Development of cybercriminal activities in the dark web has been characterized by advancement of simple drug markets to complex cybercrime ecosystems. Table II demonstrates major cybercriminal activities that were related to the dark web.

TABLE II

SIGNIFICANT DARK WEB CYBERCRIMINAL INCIDENTS

Incident	Year	Details
Silk Road Takedown	2013	FBI seized the first major dark web drug market; $1B in Bitcoin confiscated
Ashley Madison Breach	2015	37M user records leaked on dark web forums
WannaCry	2017	Ransomware spread via dark web EternalBlue exploit; $4B estimated damage
Collection #1 Breach	2019	773M email/password combinations posted on dark web
REvil RaaS	2020–2021	Ransomware-as-a-Service group sold attack kits on dark web forums
Conti Leaks	2022	Internal Conti ransomware gang chats exposed, revealing dark web operations

B. Ransomware-as-a-Service (RaaS)

Under the RaaS model, ransomware creators sell their malware infrastructure to other operators, who then carry out attacks and give a commission back to the creators, usually 2030 per cent. The most common distribution and recruitment are through dark web forums and .onion sites. RaaS groups such as REvil, Conti, LockBit and BlackCat (ALPHV) also had dedicated leaks sites on the dark web used to coerce victims by threatening to publish their data.

C. Stolen Data and Credential Markets

Markets of stolen data are some of the most widespread types of dark web threats. Collected credentials, either as a result of phishing or data breaches, or infostealer malware, are bundled and traded in large quantities. Depending on the type of data, prices will differ:

Credit card fullz (full card information): $10-50 each record.

•Corporate VPN credentials: 500-10,000 per account.

•Healthcare records: $250–$1,000 per record

•Social Security Numbers: $1-$10 per record.

This information is the ammunition of identity theft, business email compromise (BEC), and account takeover (ATO) schemes towards businesses across the globe.D. Zero-Day Exploit Markets

Zero-day vulnerabilities (vulnerabilities unknown to anyone before) are priced highly in the dark web exploit brokerages. Bids of tens of thousands to millions of dollars are made on high-value targets like enterprise software that is widely deployed, mobile operating systems, and industrial control systems. These adventures are then used to launch targeted attacks on government, financial and critical infrastructure sectors.

E. Cybercrime-as-a-Service (CaaS)

The larger Cybercrime-as-a-Service paradigm has decreased the entry entry barrier of would-be threat actors. Common services that can be easily found on the forums of the dark web are: Distributed Denial-ofservice (DDoS)-for-hire, phishing kit subscriptions, initial access brokerage, and malware development. This ecosystem allows people with little in the way of technical skills to carry out advanced attacks by buying expert services at the hands of experienced criminals.

F. Nation-State and Advanced Persistent Threat (APT) Activity

In addition to criminals driven by financial gain, the dark web also acts as a coordination platform to nation-state actors and APT groups. Often, operatives acquire bespoke equipment, find informants, and launder bitcoin funds with the help of dark web markets. Threat actors affiliated with various nation-states have been connected by intelligence agencies to procurement of zero-days and disinformation toolsing and reconnaissance data of critical infrastructure, thought to be available on dark web marketplaces.

IV. THREAT INTELLIGENCE FRAMEWORKS FOR DARK WEB MONITORING

A. The Cyber Kill Chain

The Cyber Kill Chain, developed by Lockheed Martin, is a model of the attack progression of an adversary, based on seven steps: Reconnaissance, Weaponization, Delivery, Exploitation, Installation, Command-and-Control (C2), and Actions on Objectives. DWTI adds contextual information to every step - such as finding recently added exploit kits (Weaponization) or C2 infrastructure being sold on forums (Command-and-Control).

B. MITRE ATT&CK Framework

MITRE ATT&CK framework offers a full taxonomy of adversary tactics, techniques, and procedures (TTPs). Dark web monitoring is a technique that aids ATT&CKbased analysis by matching posts to threat actors forums and samples of dark web malware with known TTP patterns. To illustrate, the sale of credential-stuffing tools are related to ATT&CK Technique T1110 (Brute Force), and initial access broker listings to T1078 (Valid Accounts).

C. Diamond Model of Intrusion Analysis

The Diamond Model provides an overview of any intrusion incident in the context of four key attributes: Adversary, Infrastructure, Capability, and Victim. DWTI implements this model through populating the Adversary and Infrastructure nodes with intelligence about the dark web: finding threat actors alias, related cryptocurrency wallets, and infrastructure (domains, IPs) promoted or discussed within criminal communities.

D. OSINT and Dark Web Threat Intelligence Integration

DWTI is based on Open Source Intelligence (OSINT) techniques. OSINT plus dark web surveillance is a structured process that takes the following form:

$$TI_{score} = w_1 \cdot C + w_2 \cdot R + w_3 \cdot T + w_4 \cdot A \quad (1)$$

TIscore is the total score of the overall threat intelligence, C is the credibility of the source, R is the relevance of the organizational assets, T is the timeliness of the intelligence, A is the actionability of the indicator and the w1,w2,w3,w4 are weighting factors that add to 1. The scoring model assists analysts in prioritizing the intelligence items obtained on the dark web.

V. MONITORING METHODOLOGIES AND TOOLS

A. Automated Dark Web Crawling

Special crawlers and scrapers are used to systematically scour through the dark web forums, marketplaces, and paste sites. In contrast to surface web crawlers, dark web crawlers have to work in Tor or I2P, have to comply with the latency limits of the anonymization network, and have to deal with dynamically generated content. OnionScan and tailor-made spiders based on Scrapy can be used in studies and corporate threat intelligence settings.

B. Keyword and Entity Monitoring

Monitoring platforms are set up by organizations to monitor particular keywords, brand names, executive names, product identifiers, and domain strings in the dark web. An alert is given when followed terms are found in new posts, allowing one to react quickly to possible data leaks, impersonation, or planning an attack.

C. Cryptocurrency Transaction Analysis

The main medium of transaction in the dark web is Bitcoin and privacy-centric cryptocurrencies like Monero, so blockchain analytics tools (Chainalysis and CipherTrace) are employed on-chain activity to track fund movements, detect wallets of threat actors, and attribute on-chain activity to real-world actors. Table III is a summary of major monitoring tools.

TABLE III

DARK WEB THREAT INTELLIGENCE TOOLS

Tool	Function
Recorded Future	Automated dark web and surface web TI aggregation
DarkOwl Vision	Dark web database indexing and entity search
Flashpoint	Criminal forum monitoring and analyst workflow
Chainalysis Reactor	Cryptocurrency transaction tracing
OnionScan	Hidden service enumeration and vulnerability scanning
Maltego	Link analysis and OSINT graph visualization
SpiderFoot	Automated OSINT collection including Tor sources

D. Human Intelligence (HUMINT) in DWTI

Although automation tools are beneficial in terms of scale, human intelligence is also essential in cases of dark web. The threat intelligence analysts who are experienced penetrate criminal forums using a disguised identity, interact with threat actors, and gather qualitative intelligence, including intended targets of planned attacks, favored tools, and internal group behaviour, which cannot be acquired by automated systems. This is what is often referred to as cyber HUMINT and should be carried out within the confines of a stringent ethical and legal framework.

E. Threat Intelligence Sharing Platforms

In order to operationalize the Intelligence of dark web at scale, organizations earnestly engage in organised sharing networks. The Malware Information Sharing Platform (MISP), Structured Threat Information Expression (STIX) and Trusted Automated Exchange of Intelligence Information (TAXII) platforms allow standardized intelligence sharing in machine-readable format across sectors, increasing the defensive value of separately-collected DWTI.

VI. EMERGING THREATS FROM THE DARK WEB

A. AI-Augmented Cybercrime

The development of large language models (LLMs) has brought about novel risks. Dark web communities of threat actors have created jailbroken or custom-built AI models - sold under brands like WormGPT and FraudGPT - specifically to help in the creation of phishing messages, malware code, and social engineering scripts without the ethical limits of larger AI systems. This ability significantly reduces the skill level to make effective assaults.

B. Initial Access Broker (IAB) Ecosystems

Initial Access Brokers do not exploit access to target networks, but sell them on dark web forums by means of compromising the networks. The IAB ecosystem has grown up considerably, and a broker provides access packages tiered, such as entry-level Remote Desktop Protocol (RDP) credentials to domain administrator access in Fortune 500 settings. IAB activity can be used as a preliminary sign of the impending ransomware deployment.

C. Critical Infrastructure Targeting

Dark web forums have seen more postings about the vulnerabilities of Industrial Control Systems (ICS) and Operational Technology (OT). Attackers have been reported to sell SCADA system credentials and reconnaissance information on water treatment plants, power plants, and factories. The integration of IT and OT networks has increased the attack surface accessible with the use of the tools purchased in the dark web.

D. Supply Chain Attack Coordination

Supply chain attacks — typified by SolarWinds, 3CX suchlike — compromise a trusted vendor to go downstream on victims. For example, dark web forums offer infrastructure for these types of attacks to coordinate their execution, anc how insiders advertise access to software development pipelines and share exploitation techniques specific to the supply chain.

VII. DEFENSIVE STRATEGIES

A. Proactive Dark Web Monitoring Programs

Organizations should also implement or buy continuous dark web monitoring with an emphasis on your particular brand, domain, executive identities and tech stack. This monitoring should take into account forums, paste sites, marketplace listings and Telegram channels associated with the dark web community. Response Window Although credential leaks can be continuously exploited by threat actors for malicious behavior, timely detection also considers the time it takes to discuss a targeted threat actor before an attack occurs.

B. Threat Intelligence-Driven Incident Response

Dark web intelligence can be part of SIEM tools and SOAR platforms. If a DWTI platform generates alerts, like corporate credentials detected on a criminal forum, automated playbooks should initiate pw changes, ajd auditing accounts and enhanced monitoring without human delay.

C. Zero Trust Architecture

In an era where credential theft and initial access brokerage targeting enterprise networks is so prevalent, I cannot recommend the Zero Trust security model highly enough. Zero Trust as the name implies never trust but always verify, requiring continuous authentication and authorization for every user, device and application with no regard to network location. This architecture reduces the value of compromised, stolen and sold credentials on the dark web.

D. Vulnerability Management and Patch Prioritization

DWTI can directly inform you about the latest vulnerabilities being exploited and sold as zero-days in dark-web markets. Dark web intelligence feeds can be incorporated into vulnerability management programs to help security teams prioritize which patches to apply first, with priority placed on the technology that appears in active exploit marketplace listings over CVSS scoring alone.

E. Employee Awareness and Insider Threat Programs

Since insider threat recruitment occurs in the dark web, with criminals advertising for employees to provide access credentials or install malware, organizations should implement active countermeasures using a competent insider threat program. The specific awareness training must include instruction on how to recognize and report attempts

to recruit externally, specifically via encrypted messaging applications.

F. Law Enforcement and Public-Private Collaboration

To properly mitigate these dark web threats, joint cooperation between the private sector and law enforcement (such as FBI Cyber Division, Europol EC3, and national CERTs) is a must. Criminals are especially vulnerable at the hands of information sharing agreements, simultaneous takedown operations (the 2022 Hydra market seizure springs to mind) and synchronised cyber intelligence shares between government and industry.

VIII. CONCLUSIONS

The dark web is an ever-present and fluid threat environment requiring dedicated collection, analytical and operational response capabilities. We have explored the infrastructure supporting the dark web, outlined primary classes of cybercriminal activities, and analyzed existing frameworks for conducting threat intelligence on the dark web. As new threats such as AIaugmented cybercrime, initial access brokerage and critical infrastructure targeting continue to undermine security operations around the world, dark web threat intelligence has become urgent for organizations that want to harden their defenses against these threats.

Suspicious behaviour, and incident analysis including proactive monitoring, Zero Trust architecture, threat intelligence-driven incident response, and law enforcement collaboration are all layers in an overall approach to mitigating dark web-sourced risk. The dark web keeps evolving and so must the intel pathologies and defensive technologies used against it. Future work should focus on standardisation of DWTI collection

APPENDIX

You have been told how the layered architecture of the internet and where the dark web fits in it. Various categories of cybercriminal activities, threat intelligence frameworks, monitoring methodologies and their associated tooling have been studied. New threats such as AI-augmented cybercrime and initial access brokerage ecosystems have also been reported. Defense in depth has been proposed across technical, procedural, and collaborative elements.

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